2 — Divergence

$$\nabla \cdot \vec{v} = \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right)$$

Analogous to → **Fluids** → $\nabla \cdot \vec{v} = 0$

if fluid in = fluid out in a particular volume. ⟹ incompressible fluid (Analogy.) / incompressible phase flow (Velocity field here)

Minus First Law (Susskind's)

ie. satisfies:

Liouville's Theorem

This incompressibility of the phase space fluid is called: Liouville's Theorem

no. of coord = N ⟹ 2N dimensional phase space (p_i, q_i)

⟹ 2N components of $\vec{v}$ field (v_{p_i}, v_{q_i}) → $(\dot{p}_i, \dot{q}_i)$ → $\left(-\frac{\partial H}{\partial q_i}, \frac{\partial H}{\partial p_i}\right)$ (from H's eq.ⁿ)

$$\nabla \cdot \vec{v} = \sum_i \left(\frac{\partial v_{q_i}}{\partial q_i} + \frac{\partial v_{p_i}}{\partial p_i}\right) = 0 \quad \text{(if fluid is incompressible)}$$

$$\nabla \cdot \vec{v} = \sum_i \left(\frac{\partial}{\partial q_i}\frac{\partial H}{\partial p_i} + \frac{\partial}{\partial p_i}\frac{\partial H}{\partial q_i}\right)$$

1 — Derive H

$\mathcal{L}(q_i, \dot{q}_i, t)$ ← explicit

$$\frac{d\mathcal{L}}{dt} = \left(\frac{\partial \mathcal{L}}{\partial q_i}\dot{q}_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\ddot{q}_i\right) + \frac{\partial \mathcal{L}}{\partial t} \quad (\to 0 \text{ if implicit})$$

$$= \dot{p}_i\dot{q}_i + p_i\ddot{q}_i$$

$$\frac{d\mathcal{L}}{dt} = \frac{d}{dt}(p_i\dot{q}_i)$$

$$\frac{d}{dt}(\mathcal{L} - p_i\dot{q}_i) = 0$$

$$\boxed{\mathcal{L} - p_i\dot{q}_i = \text{const}} = H$$

$$\tfrac{1}{2}m\dot{q}_i^2 - V(q_i) - m\dot{q}_i^2$$

$$-\left(\tfrac{1}{2}m\dot{q}_i^2 + V(q_i)\right) = \text{const.}$$

aka. E_s = const

$$H = \sum_i \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \mathcal{L}$$

Derive Hamilton's Eq.ⁿ

$$H = \sum_i (p_i\dot{q}_i) - \mathcal{L}(q_i, \dot{q}_i)$$

$$\delta H = \sum_i \left(\dot{q}_i\delta p_i + p_i\delta\dot{q}_i - \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i - \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\delta\dot{q}_i\right)$$

$$\delta H = \sum_i \left(\dot{q}_i\delta p_i - \frac{\partial \mathcal{L}}{\partial q_i}\delta q_i\right) \quad \text{— (i)} \qquad \left(\frac{\partial \mathcal{L}}{\partial q_i} \to \dot{p}_i\right)$$

$$\delta H(q_i, p_i) = \sum_i \left(\frac{\partial H}{\partial p_i}\delta p_i + \frac{\partial H}{\partial q_i}\delta q_i\right) \quad \text{— (ii)}$$

Comparing (i) ; (ii) :

$$\boxed{\frac{\partial H}{\partial p_i} = \dot{q}_i \; ; \; \frac{\partial H}{\partial q_i} = -\dot{p}_i}$$

Example - 1.1

→ $\dot{q} = \sqrt[4]{km}\,\dot{x}$

⟹ $\dot{q}^4 = km\dot{x}^4$

⟹ $\dot{x}^2 = \frac{\dot{q}^2}{\sqrt{km}}$

$\omega = \sqrt{\frac{K}{m}}$

$$\mathcal{L} = \frac{m\dot{x}^2}{2} - \frac{kx^2}{2}$$

$$= \frac{\dot{q}^2}{2\omega} - \frac{\omega}{2}q^2$$

$$\boxed{\mathcal{L} = \frac{\dot{q}^2}{2\omega} - \frac{\omega q^2}{2}}$$

Example - 1.2 convert to H

$$\mathcal{L} = \frac{\dot{q}^2}{2\omega} - \frac{\omega}{2}q^2$$

$$H = p\dot{q} - \mathcal{L}$$

$$= p\dot{q} - \frac{\dot{q}^2}{2\omega} + \frac{\omega}{2}q^2$$

$$H = \frac{\dot{q}^2}{\omega} - \frac{\dot{q}^2}{2\omega} + \frac{\omega}{2}q^2$$

$$= \frac{\omega\dot{q}^2}{2\omega^2} + \frac{\omega}{2}q^2$$

$$H = \frac{p^2\omega}{2} + \frac{q^2\omega}{2} \quad \left[\because p = \frac{\dot{q}}{\omega}\right]$$

$$H = \frac{\omega}{2}(p^2 + q^2)$$

Poisson's Bracket

$$\dot{F}(q_i, p_i) = \sum_i \frac{\partial F}{\partial q_i}\dot{q}_i + \frac{\partial F}{\partial p_i}\dot{p}_i$$

$$\dot{F}(q_i, p_i) = \sum_i \frac{\partial F}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial H}{\partial q_i}$$

$$\boxed{\dot{F} = \{F, H\}} \quad \text{\{Poisson's Bracket\}}$$

i.e.

$$\{F, G\} = \sum_i \left(\frac{\partial F}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q_i}\right)$$

Rules :

i) $\{F, G\} = -\{G, F\}$

ii) $\{kF, G\} = k\{F, G\}$ ($k \equiv$ const.)

iii) $\{(A+F), G\} = \{A, G\} + \{F, G\}$

iv) $\{(AF), G\} = F\{A, G\} + A\{F, G\}$

v) $\{q_i, p_j\} = \delta_{ij} = \begin{cases} 0 ; & i \neq j \\ 1 ; & i = j \end{cases}$

???

$$\dot{F} = \left\{F(q,p), \dot{q}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \mathcal{L}\right\}$$

$$\dot{F} = \left\{F(q,p), \dot{q}\frac{\partial \mathcal{L}}{\partial \dot{q}}\right\} - \{F(q,p), \mathcal{L}\}$$

$$\dot{F} = \dot{q}\left\{F(q,p), \frac{\partial \mathcal{L}}{\partial \dot{q}}\right\} + \frac{\partial \mathcal{L}}{\partial \dot{q}}\{F(q,p), \dot{q}\} - \{F(q,p), \mathcal{L}\}$$

What now ? Think !!! (Trying to make something new.)

Proofs :

i) $\{F, G\} = \sum_i \left(\frac{\partial F}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q_i}\right)$

$\{G, F\} = \sum_i \left(\frac{\partial G}{\partial q_i}\frac{\partial F}{\partial p_i} - \frac{\partial G}{\partial p_i}\frac{\partial F}{\partial q_i}\right) = -\{F, G\}$

Hence Proved !

ii) $\{kF, G\} = \sum_i \left(k\frac{\partial F}{\partial q_i}\frac{\partial G}{\partial p_i} - k\frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q_i}\right)$

$\{kF, G\} = k\sum_i \left(\frac{\partial F}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q_i}\right)$

$\{kF, G\} = k\{F, G\}$

Hence Proved !

iii) $\{(A+F), G\} = \sum_i \left(\frac{\partial(A+F)}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial(A+F)}{\partial p_i}\frac{\partial G}{\partial q_i}\right)$

$= \sum_i \left(\left(\frac{\partial A}{\partial q_i} + \frac{\partial F}{\partial q_i}\right)\frac{\partial G}{\partial p_i} - \left(\frac{\partial A}{\partial p_i} + \frac{\partial F}{\partial p_i}\right)\frac{\partial G}{\partial q_i}\right)$

$\{(A+F), G\} = \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial G}{\partial p_i} + \frac{\partial F}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial G}{\partial q_i} - \frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q_i}\right)$

$= \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial G}{\partial q_i}\right) + \sum_i \left(\frac{\partial F}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q_i}\right)$

$\{(A+F), G\} = \{A, G\} + \{F, G\}$

Hence Proved !

iv) $\{(AF), G\} = \sum_i \left(\frac{\partial(AF)}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial(AF)}{\partial p_i}\frac{\partial G}{\partial q_i}\right)$

$= \sum_i \left(\left[F\frac{\partial A}{\partial q_i} + A\frac{\partial F}{\partial q_i}\right]\frac{\partial G}{\partial p_i} - \left[F\frac{\partial A}{\partial p_i} + A\frac{\partial F}{\partial p_i}\right]\frac{\partial G}{\partial q_i}\right)$

$\{(AF), G\} = \sum_i \left(\left[F\frac{\partial A}{\partial q_i}\frac{\partial G}{\partial p_i} + A\frac{\partial F}{\partial q_i}\frac{\partial G}{\partial p_i}\right] - \left[F\frac{\partial A}{\partial p_i}\frac{\partial G}{\partial q_i} + A\frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q_i}\right]\right)$

$= \sum_i \left(F\left[\frac{\partial A}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial G}{\partial q_i}\right] + A\left[\frac{\partial F}{\partial q_i}\frac{\partial G}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial G}{\partial q_i}\right]\right)$

$\{(AF), G\} = F\{A, G\} + A\{F, G\}$

Hence Proved !

(notation problem ?)

v) $\{q_i, p_j\}$: if $i = j$

$= \sum_i \frac{\partial q_i}{\partial q_i}\frac{\partial p_i}{\partial p_i} - \frac{\partial q_i}{\partial p_i}\frac{\partial p_i}{\partial q_i}$

$= 1$

if $i \neq j$

$= \sum_i \frac{\partial q_i}{\partial q_i}\frac{\partial p_j}{\partial p_i} - \frac{\partial q_i}{\partial p_i}\frac{\partial p_j}{\partial q_i}$

$= 0$

Hence Proved !

Example - 2

$$H = \frac{p^2}{2m} + \frac{k}{2}q^2$$

$\dot{q} = \{q, H\} = \left\{q, \frac{p^2}{2m} + \frac{k}{2}q^2\right\}$

$= \left[\left\{q, \frac{p^2}{2m}\right\} + \left\{q, \frac{k}{2}q^2\right\}\right]$ (→ 0)

$\dot{q} = \left\{q, \frac{p^2}{2m}\right\} = \frac{1}{2m}\{q, p^2\}$

$= \frac{1}{2m}\{q, p(p)\}$

$\dot{q} = \frac{1}{2m}\left[p\{q, p\} + p\{q, p\}\right]$

$\dot{q} = \frac{p}{m}\{q, p\}$ (→ 1)

$\dot{q} = \frac{p}{m}$ — (i)

$\dot{p} = \{p, H\} = \left\{p, \frac{p^2}{2m} + \frac{k}{2}q^2\right\}$

$= \left\{p, \frac{k}{2}q^2\right\}$

$\dot{p} = \frac{k}{2}\{p, q(q)\}$

$= \frac{k}{2}\left[q\{p, q\} + q\{p, q\}\right]$

$\dot{p} = kq\{p, q\}$ (→ −1)

$\dot{p} = -kq$ — (ii)

$\frac{d}{dt}$(i) ⟹ $\ddot{q} = \frac{\dot{p}}{m}$

from (ii) ⟹ $\ddot{q} = -\frac{kq}{m}$

⟹ $\boxed{m\ddot{q} = -kq}$

(Newton's 2nd law for harmonic oscillator)